

SIDDHIVINAYAK RAGHAVRAJU

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Denton, TX

PROFESSIONAL SUMMARY

Full Stack Developer with 2+ years of professional experience building scalable, production-grade web applications, combined with graduate-level expertise in Artificial Intelligence and Machine Learning. Proven ability to deliver end-to-end systems using React, Node.js, and AWS, and to apply advanced ML techniques including adversarial NLP and transformer fine-tuning. MS thesis research demonstrates BERT accuracy collapsing from **86% to 12%** under adversarial attack — directly relevant to safe AI deployment. Seeking full-time roles in Full Stack Development or AI/ML Engineering.

EDUCATION

University of North Texas

Denton, TX

Master of Science in Artificial Intelligence

Aug. 2024 – Dec. 2026

- **Thesis:** Adversarial Robustness of Temporal Relation Classifiers — fine-tuning BERT and RoBERTa on the TLINK dataset, evaluated against 15 adversarial NLP attacks (TextFooler, PWWS, BAE, BERT-Attack, DeepWordBug, and more). Key finding: model accuracy collapsed **86% → 12%** under PWWS with only ~10% word substitution.
- **Coursework:** Deep Learning, Advanced Machine Learning, Computer Vision, Analysis of Algorithms

TECHNICAL SKILLS

Languages: Python, JavaScript

Frontend & Backend: React.js, Node.js, Express.js

AI / ML Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers

AI Specializations: NLP, Adversarial Machine Learning, CNNs, RNNs, LSTMs, BERT, GPT, Transfer Learning

Cloud & DevOps: AWS, Docker, Git, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

EXPERIENCE

Full Stack Software Developer

Dec. 2022 – Aug. 2024

Infynect Labs Private Limited

Bangalore, India

- Built and maintained end-to-end web applications for a live IPTV streaming platform serving thousands of concurrent users, using **React.js**, **Node.js**, and **MySQL**.
- Integrated **Digital Rights Management (DRM)** solutions ensuring secure, licensed content delivery — a compliance-critical requirement in the media streaming industry.
- Containerized microservices with **Docker** and deployed on **AWS**, improving deployment reliability and reducing release cycle time.
- Designed and implemented workflow automation pipelines, increasing team productivity by **15%** and reducing manual engineering overhead.
- Collaborated with product, design, and QA teams in an agile environment to ship features on time and aligned with business goals.

PROJECTS

Adversarial Robustness of NLP Models | *Python, PyTorch, HuggingFace, TextAttack* | 🐙 github.com/vinayakraju16/Adversarial-ML

- Fine-tuned **BERT** and **RoBERTa** on the TLINK temporal relation classification dataset and evaluated robustness against **15 adversarial NLP attacks** including TextFooler, PWWS, BAE, and BERT-Attack.
- Demonstrated accuracy collapse from **86% → 12%** under PWWS attack with ~10% word substitution; overall attack success rate of **86%**.
- Built a fully config-driven pipeline (YAML) with GPU-accelerated training, multi-attack evaluation, and automated results logging for reproducible research.

Credit Card Fraud Detection System | *Python, Scikit-learn, Flask, REST API*

- Developed a logistic regression ML pipeline achieving **87% accuracy** in fraud detection, with REST API endpoints and a real-time Flask dashboard for predictions.
- Reduced false positives by **16%** via feature engineering and hyperparameter tuning; improved model efficiency by **23%** using ensemble and under-sampling techniques.

Early Heart Disease Prediction | *Python, Scikit-learn, Matplotlib, Tableau*

- Built a HIPAA-compliant predictive ML model improving early heart disease detection outcomes by **26%** through accurate patient risk factor classification.
- Implemented end-to-end data encryption protocols reducing data breach risk by **40%**, with data pipelines and dashboards built in Matplotlib and Tableau.